

Gauß-Newton

Bobitsmagic

$$f(x, y) = \sum_i^P x_i y_i$$

$$E(x) = \frac{1}{2} \sum_s^S (f(x, p_s) - t_s)^2$$

$$\Delta E = \begin{pmatrix} \frac{\delta E}{\delta x_0} \\ \frac{\delta E}{\delta x_1} \\ \vdots \end{pmatrix}$$

$$\Delta_E = \begin{pmatrix} \sum_s^S (f(x, p^s) - t^s)(p_0^s) \\ \sum_s^S (f(x, p^s) - t^s)(p_1^s) \\ \vdots \end{pmatrix}$$

$$H_E = \begin{pmatrix} \sum (p_0^s)^2 & \sum p_0^s p_1^s & \dots \\ \sum p_1^s p_0^s & \sum (p_1^s)^2 & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

$$H_E v = \Delta_E$$

$$E(x) = \sum_s \left(\left(\sum_p x_p y_{sp} \right) - t_s \right)^2$$

$$\frac{1}{2} \left(f_{s(x)} \right)^2 = \left(\left(\sum_p x_p y_{sp} \right) - t_s \right)^2$$

$$f_{s(x)} = \sqrt{2} \left(\left(\sum_p x_p y_{sp} \right) - t_s \right)$$

$$J = \begin{pmatrix} \frac{\delta f_0}{\delta x_0} & \frac{\delta f_0}{\delta x_1} & \dots \\ \frac{\delta f_1}{\delta x_0} & \frac{\delta f_1}{\delta x_1} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

$$J = \sqrt{2} \begin{pmatrix} y_{00} & y_{01} & \dots \\ y_{10} & y_{11} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

$$J^T J = (y_{00}^2)$$

1 Move generation

1.1 Pseudo moves

Pawns: Bitboard parallel Knights: Per knight Slider: Per slider perfect hashing or bitmask sort or

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} (1 \ 0) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} (c \ d) = \begin{pmatrix} ac & ad \\ bc & bd \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \rightarrow \begin{pmatrix} a & a & a \\ b & b & b \\ c & c & c \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow \begin{pmatrix} x & y & z \\ x & y & z \\ x & y & z \end{pmatrix}$$

$$M = \begin{pmatrix} a & a & a \\ b & b & b \\ c & c & c \end{pmatrix} + \begin{pmatrix} x & y & z \\ x & y & z \\ x & y & z \end{pmatrix}$$

$$||P-M||_2^2$$

$$\sum_{i,j} (P_{ij} - M_{ij})^2$$

$$\sum_{i,j} (P_{ij} - (c_i + r_j))^2$$

$$\begin{aligned} \frac{\partial}{\partial c_k} \sum_{i,j} (P_{ij} - (c_i + r_j))^2 &= \frac{\partial}{\partial c_k} \sum_j^n (P_{kj} - (c_k + r_j))^2 \\ &= \sum_j^n \frac{1}{2} (P_{kj} - (c_k + r_j)) \cdot (-1) \\ &= -\frac{1}{2} \sum_j^n (P_{kj} - (c_k + r_j)) \\ &= -\frac{1}{2} \sum_j^n P_{kj} - (c_k + r_j) \\ &= -\frac{1}{2} \left(-nc_k + \sum_j P_{kj} - r_j \right) \\ &= \frac{1}{2} \left(nc_k - \sum_j P_{kj} - r_j \right) \\ &\rightarrow nc_k + \sum_j r_j = \sum_j P_{kj} \\ &\rightarrow nr_k + \sum_i c_i = \sum_i P_{ik} \end{aligned}$$

1.2 Example

$$n = 2$$

$$P = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$\begin{pmatrix} c_0 \\ c_1 \end{pmatrix}, (r_0 \ r_1)$$

$$\begin{pmatrix} n & 0 & 1 & 1 \\ 0 & n & 1 & 1 \\ 1 & 1 & n & 0 \\ 1 & 1 & 0 & n \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ r_0 \\ r_1 \end{pmatrix} = \begin{pmatrix} \sum_j P_{0j} \\ \sum_j P_{1j} \\ \sum_i P_{i0} \\ \sum_i P_{i1} \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 2 & 0 \\ 1 & 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ r_0 \\ r_1 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \\ 4 \\ 6 \end{pmatrix}$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 1 & 7 \\ 1 & 1 & 2 & 0 & 4 \\ 1 & 1 & 0 & 2 & 6 \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 1 & 7 \\ 2 & 2 & 4 & 0 & 8 \\ 2 & 2 & 0 & 4 & 12 \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 1 & 7 \\ 0 & 2 & 3 & -1 & 5 \\ 0 & 2 & -1 & 3 & 9 \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 1 & 7 \\ 0 & 0 & 2 & -2 & -2 \\ 0 & 0 & -2 & 2 & 2 \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 1 & 7 \\ 0 & 0 & 2 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 1 & 7 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$r_0 = r_1 - 1$$

$$2c_1 + r_0 + r_1 = 7 \Leftrightarrow c_1 = 4 - r_1$$

$$2c_0 + r_0 + r_1 = 3 \Leftrightarrow c_0 = 2 - r_1$$

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ r_0 \\ r_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 1 & 4 \end{array} \right)$$

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$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 3 \\ 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$r_1 - 1 = r_0$$

$$c_1 + r_0 = 3 \Leftrightarrow c_1 = 4 - r_1$$

$$c_0 + r_0 = 1 \Leftrightarrow c_0 = 2 - r_1$$

$$(-1 \ 0) + \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ r_0 \\ r_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & a \\ 1 & 0 & 0 & 1 & b \\ 0 & 1 & 1 & 0 & c \\ 0 & 1 & 0 & 1 & d \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & a \\ 0 & 1 & 1 & 0 & c \\ 0 & 0 & -1 & 1 & b-a \\ 0 & 1 & 0 & 1 & d \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & a \\ 0 & 1 & 1 & 0 & c \\ 0 & 0 & -1 & 1 & b-a \\ 0 & 0 & -1 & 1 & d-c \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & a \\ 0 & 1 & 1 & 0 & c \\ 0 & 0 & -1 & 1 & b-a \\ 0 & 0 & 0 & 0 & d-c-b+a \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & a \\ 0 & 1 & 1 & 0 & c \\ 0 & 0 & -1 & 1 & b-a \\ 0 & 0 & 0 & 0 & d-c-b+a \end{array} \right)$$

$$r_1 - 1 = r_0$$

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$$(-1 \ 0) + \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$n = 2$$

$$P = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\begin{pmatrix} c_0 \\ c_1 \end{pmatrix}, (r_0 \ r_1)$$

$$\begin{pmatrix} n & 0 & 1 & 1 \\ 0 & n & 1 & 1 \\ 1 & 1 & n & 0 \\ 1 & 1 & 0 & n \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ r_0 \\ r_1 \end{pmatrix} = \begin{pmatrix} \sum_j P_{0j} \\ \sum_j P_{1j} \\ \sum_i P_{i0} \\ \sum_i P_{i1} \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 2 & 0 \\ 1 & 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ r_0 \\ r_1 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \\ 4 \\ 6 \end{pmatrix}$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & a+b \\ 0 & 2 & 1 & 1 & c+d \\ 1 & 1 & 2 & 0 & a+c \\ 1 & 1 & 0 & 2 & b+d \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & a+b \\ 0 & 2 & 1 & 1 & c+d \\ 2 & 2 & 4 & 0 & 2a+2c \\ 2 & 2 & 0 & 4 & 2b+2d \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & a+b \\ 0 & 2 & 1 & 1 & c+d \\ 0 & 2 & 3 & -1 & a-b+2c \\ 0 & 2 & -1 & 3 & -a+b+2d \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & a+b \\ 0 & 2 & 1 & 1 & c+d \\ 0 & 0 & 2 & -2 & a-b+c-d \\ 0 & 0 & -2 & 2 & -a+b-c+d \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 1 & 7 \\ 0 & 0 & 2 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 1 & 7 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$r_0 = r_1 + \frac{a-b+c-d}{2}$$

$$2c_1 + r_0 + r_1 = c + d \Leftrightarrow c_1 = -\frac{a-b+c-d}{4} - r_1 + \frac{c+d}{2}$$

$$2c_0 + r_0 + r_1 = 3 \Leftrightarrow c_0 = -\frac{a-b+c-d}{4} - r_1 + \frac{a+b}{2}$$

$$\begin{aligned}
\frac{\partial}{\partial c_k} \sum_{i,j} (P_{ij} - (c_i \cdot r_j))^2 &= \frac{\partial}{\partial c_k} \sum_j^n (P_{kj} - (c_k \cdot r_j))^2 \\
&= \sum_j^n 2(P_{kj} - (c_k \cdot r_j)) \cdot (-r_j) \\
&= -2 \sum_j^n (P_{kj} - (c_k \cdot r_j)) \cdot r_j \\
&= -2 \sum_j^n P_{kj} r_j - c_k \cdot r_j^2 \\
&\rightarrow \sum_j^n P_{kj} r_j - c_k \sum_j^n r_j^2
\end{aligned}$$